FOUR EQUIVALENT CHARACTERIZATIONS OF VERSALITY AT A FINITE TYPE POINT; THE COMPLETED LOCAL RING IS A VERSAL RING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 0DQU: sources/stacks-chow/tags.txt, source lines 17199–17199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07XF: sources/stacks-chow/tags.txt, source lines 10075–10075. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0CXK: sources/stacks-chow/tags.txt, source lines 16210–16210. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07WK: sources/stacks-chow/tags.txt, source lines 10045–10045. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06EJ: sources/stacks-chow/tags.txt, source lines 8218–8218. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0CXM: sources/stacks-chow/tags.txt, source lines 16212–16212. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0DQU let $x : U \to \mathcal{X}$ be a morphism where U is a scheme locally of finite type over S. Let $u_0 \in U$ be a finite type point. Set $k = \kappa(u_0)$ and denote $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ the induced map. The following are equivalent*

1. *x is versal at u_0 (Artin's Axioms, Definition 07XF),*
2. *$\hat{x} : \mathcal{F}_{U,k,u_0} \to \mathcal{F}_{\mathcal{X},k,x_0}$ is smooth,*
3. *the formal object associated to $x|_{\mathrm{Spec}(\mathcal{O}^\wedge_{U,u_0})}$ is versal, and*

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(4) *there is an open neighbourhood $U' \subset U$ of x such that $x|_{U'} : U' \to \mathcal{X}$ is smooth.*

Moreover, in this case the completion $\mathcal{O}^\wedge_{U,u_0}$ is a versal ring to $\mathcal{X}$ at x_0.

Proof. Since $U \to S$ is locally of finite type (as a composition of such morphisms), we see that $\mathrm{Spec}(k) \to S$ is of finite type (again as a composition). Thus the statement makes sense. The equivalence of (1) and (2) is the definition of x being versal at u_0. The equivalence of (1) and (3) is Artin's Axioms, Lemma 0CXK. Thus (1), (2), and (3) are equivalent.

If $x|_{U'}$ is smooth, then the functor $\hat{x} : \mathcal{F}_{U,k,u_0} \to \mathcal{F}_{\mathcal{X},k,x_0}$ is smooth by Artin's Axioms, Lemma 07WK. Thus (4) implies (1), (2), and (3). For the converse, assume x is versal at u_0. Choose a surjective smooth morphism $y : V \to \mathcal{X}$ where V is a scheme. Set $Z = V \times_{\mathcal{X}} U$ and pick a finite type point $z_0 \in |Z|$ lying over u_0 (this is possible by Morphisms of Spaces, Lemma 06EJ). By Artin's Axioms, Lemma 0CXM the morphism $Z \to V$ is smooth at z_0. By definition we can find an open neighbourhood $W \subset Z$ of z_0 such that $W \to V$ is smooth. Since $Z \to U$ is open, let $U' \subset U$ be the image of W. Then we see that $U' \to \mathcal{X}$ is smooth by our definition of smooth morphisms of stacks.

The final statement follows from the definitions as $\mathcal{O}^\wedge_{U,u_0}$ prorepresents $\mathcal{F}_{U,k,u_0}$. $\square$